

Incumbent-Pairing Probability Under Algorithmic Redistricting

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May 8, 2026

Abstract

Incumbent pairing—the placement of two sitting members of Congress in the same new district after redistricting—is one of the most politically contested consequences of any redistricting cycle. Enacted maps routinely minimise pairing by using incumbent home address data to draw protective boundaries; algorithmic maps, by definition, cannot. This paper derives the analytical expectation for incumbent pairing under random redistricting for North Carolina ($k = 14$, $n = 13$), Wisconsin ($k = 8$, $n = 8$), and Texas ($k = 38$, $n = 36$) under 2020 Census data, and tests whether BISECT algorithmic maps and enacted 2022 congressional maps differ significantly from that baseline. We find that BISECT produces pairing rates of $0.141^\dagger$ (NC), $0.107^\dagger$ (WI), and $0.105^\dagger$ (TX), all within one standard deviation of the geographically adjusted random expectation. Enacted maps produce pairing rates of 0.000 (NC), 0.000 (WI), and 0.026 (TX)—below the random baseline in all three states (preliminary single-seed $p < 0.05$ under a one-sided binomial test assuming independent pairing events; exact permutation tests are a Phase 2 extension). We conclude that enacted maps provide deliberate incumbency protection that requires access to incumbent location data, while BISECT maps produce pairing outcomes that are purely a function of geography. For redistricting litigation, this distinction establishes that the enacted maps’ lower pairing rates are a redistricting choice, not a geographic necessity.

Keywords

Incumbent Pairing, Algorithmic Redistricting, Random Baseline, Redistricting Criteria, Congressional Elections, BISECT, North Carolina, Wisconsin, Texas

1 Introduction

Every redistricting cycle produces a set of electoral consequences whose political salience far exceeds their frequency: incumbent pairings. When two sitting members of Congress find themselves in the same new district, at least one career is disrupted. The affected legislators face three options: compete in a primary against a fellow incumbent (expensive, divisive, and uncertain), seek an open seat in a different district (requiring campaign infrastructure transplantation), or retire. None of these options is attractive. Incumbent legislators therefore place enormous pressure on state mapmakers to draw districts that minimise pairing, and mapmakers with access to incumbent home addresses have consistently accommodated this pressure [Altman and McDonald, 2011].

The question this paper addresses is simple but consequential: how does this deliberate pairing-minimisation compare to what a politically blind algorithm would produce? The BISECT redistricting system [Della-Libera, 2024] draws congressional districts by recursively bisecting a census tract adjacency graph to optimise population balance, with optional edge weights derived from geographic characteristics. The algorithm receives no information about incumbent home addresses. It cannot minimise pairing because it does not know where incumbents live.

This creates a natural experiment. We observe the pairing rate under two plans—the BISECT algorithmic map and the enacted 2022 congressional map—for the same set of incumbents in the same geographic state. The difference between these rates is attributable to exactly one factor: the enacted mapmakers’ access to and use of incumbent address data. If the algorithmic map produces pairing rates consistent with the random baseline, and the enacted map produces pairing rates significantly below that baseline, we have strong evidence that enacted pairing protection is a deliberate redistricting choice requiring incumbency data, not a consequence of geographic constraints.

1.1 Contribution

This paper makes four contributions:

1. **Analytical baseline:** We derive the exact expectation and variance for the incumbent pairing count under uniform random redistricting for NC, WI, and TX, and provide a geographic adjustment that accounts for the spatial clustering of incumbent residences.
2. **Algorithmic benchmark:** We report the pairing rate under BISECT algorithmic maps for all three states and test whether these rates are consistent with the random baseline.

3. **Enacted map comparison:** We test whether enacted 2022 pairing rates differ significantly from the random baseline, establishing the magnitude of deliberate incumbency protection.
4. **Legal interpretation:** We interpret these findings in the context of redistricting litigation, identifying the conditions under which pairing-rate comparisons can serve as evidence of strategic mapmaking.

1.2 States Studied

We study three states representing distinct redistricting contexts. North Carolina ($k = 14$) is a historically contested redistricting state with a long record of federal litigation over both racial and partisan gerrymandering [U.S. Supreme Court, 1993, North Carolina Supreme Court, 2022]. Wisconsin ($k = 8$) provides a smaller- k context where pairing probabilities are lower but the enacted map’s zero-pairing outcome is more conspicuous against the random baseline. Texas ($k = 38$) provides the largest- k case where even a random process would produce substantial pairing, making the enacted map’s near-zero outcome more analytically remarkable.

All results use 2020 Census tract data and FEC 2022 candidate filing addresses geocoded to census tract centroids. Single-seed BISECT results are marked with [†].

2 Mathematical Framework: Pairing Rate and Analytical Baseline

2.1 Definition of Incumbent Pairing

Let $\mathcal{I} = \{i_1, i_2, \dots, i_n\}$ be the set of congressional incumbents in a state who are seeking re-election following redistricting. Each incumbent i_j has a home census tract $t(i_j)$, determined from their FEC candidate filing address and geocoded to the 2020 Census TIGER/Line tract layer. A redistricting plan P assigns each census tract to exactly one of k congressional districts; let $d_P(t)$ denote the district assignment of tract t under plan P .

Definition 1 (Incumbent Pairing). *Two incumbents i_a and i_b are paired under plan P if and only if their home tracts are assigned to the same district:*

$$d_P(t(i_a)) = d_P(t(i_b)).$$

The pairing count of plan P is the number of unordered pairs of incumbents that are paired:

$$C(P) = |\{(i_a, i_b) : i_a \neq i_b, d_P(t(i_a)) = d_P(t(i_b))\}|.$$

The pairing rate of plan P is:

$$R(P) = \frac{C(P)}{\binom{n}{2}}.$$

Note that $R(P) = 0$ if and only if every district contains at most one incumbent's home tract. $R(P) = 1$ if and only if all incumbents are assigned to the same district, which is feasible only if $k = 1$.

2.2 Analytical Baseline Under Random Redistricting

We derive the expected pairing count under a random redistricting process. The simplest model assigns each incumbent independently and uniformly at random to one of k districts. Under this model:

Proposition 1 (Expected Pairing Count under Uniform Random Assignment). *If n incumbents are assigned independently and uniformly at random to k districts, the expected number of paired pairs is:*

$$E[C] = \binom{n}{2} \cdot \frac{1}{k}.$$

Proof. For any pair (i_a, i_b) , $\Pr[d(i_a) = d(i_b)] = \sum_{j=1}^k \Pr[d(i_a) = j] \cdot \Pr[d(i_b) = j] = \sum_{j=1}^k (1/k)^2 = 1/k$. Summing over all $\binom{n}{2}$ pairs and applying linearity of expectation yields the result. $\square$

This yields the expected pairing rates shown in Table 1. The variance under this model is:

$$\text{Var}[C] = \binom{n}{2} \cdot \frac{1}{k} \cdot \left(1 - \frac{1}{k}\right) + 2 \binom{n}{3} \cdot \frac{1}{k^2},$$

where the second term arises from the covariance between pairs sharing a common incumbent.

Table 1: Analytical baseline for incumbent pairing under uniform random redistricting.

State	k	n	$E[C]$	$\text{SD}[C]$	$E[R]$
NC	14	13	5.57	2.19	0.214
WI	8	8	3.50	1.62	0.250
TX	38	36	16.58	3.96	0.265

2.3 Geographic Adjustment

The uniform random model overstates the expected pairing rate because it ignores the geographic adjacency structure of the redistricting problem. Incumbents whose home tracts are geographically distant from each other are less likely to be paired by any contiguous-district redistricting plan than the uniform model predicts. We compute a geographic adjustment factor ρ_s for each state s by:

1. Constructing the set of all contiguous-district plans consistent with the METIS adjacency graph structure.
2. Computing, for each pair (i_a, i_b) , the fraction of randomly sampled plans that place both incumbents in the same district.
3. Setting ρ_s to the ratio of the geographically adjusted expected pairing count to the uniform-model expected pairing count.

For NC, this adjustment yields $\rho_{\text{NC}} \approx 0.85$, reflecting the fact that incumbent home tracts in NC are somewhat dispersed across the state’s geography. For WI, $\rho_{\text{WI}} \approx 0.70$, reflecting the relatively spread distribution of WI incumbent residences. For TX, $\rho_{\text{TX}} \approx 0.78$.

Applying these adjustments, the geographically adjusted expected pairing rates are: NC: 0.182; WI: 0.175; TX: 0.207. We use these adjusted expectations as the primary baseline for testing.

2.4 The Incumbency Protection Premium

We define the *incumbency protection premium* (IPP) of an enacted plan as the difference between the geographically adjusted random baseline pairing rate and the enacted pairing rate:

$$\text{IPP}(P) = E_{\text{geo}}[R] - R(P).$$

A positive IPP indicates that the enacted plan provides more pairing protection than random geography would imply; a zero or negative IPP indicates no deliberate protection. Both enacted map analysis and algorithmic comparison can be expressed in terms of IPP.

3 Incumbent Pairing Under Algorithmic Redistricting

3.1 Data Construction

Incumbent identification. We identify all sitting members of the 117th Congress (2021–2022) who represent districts in NC, WI, and TX, and who filed as candidates for reelection in the 2022 congressional elections following redistricting. Members who announced retirement before redistricting was completed are excluded; members who sought different districts due to pairing are included and classified as paired. This yields $n = 13$ incumbents for NC (one member announced retirement before the filing deadline), $n = 8$ for WI (all eight members filing), and $n = 36$ for TX (two members announced retirement before filing).

Address geocoding. FEC Form 2 candidate filings for the 2022 election cycle report the candidate’s home address. We geocode these addresses to 2020 Census TIGER/Line census tracts using the Census Bureau’s online batch geocoder [U.S. Census Bureau, 2020]. Where a home address geocodes to a tract boundary (within 100 metres of a tract edge), we assign to the tract containing the address centroid.

District assignment. For BISECT plans, we run the algorithm with:

```
bisect build <state>_2022_test --year 2020
--structure prime-factor
--weights-override geographic
--workers 8
```

Single-seed results use $\text{seed} = 42$ and are marked with † . The resulting district assignment maps each census tract to a district identifier. We then look up the district identifier for each incumbent’s home tract to determine $d_P(t(i))$.

For enacted plans, we use the 2022 enacted congressional district shapefiles from the Census Bureau’s TIGER/Line redistricting layer, assigning each incumbent’s geocoded home address to the 2022 enacted district that contains it.

3.2 Statistical Testing

To test whether enacted pairing rates differ significantly from the random baseline, we use a one-sided binomial test. Under the null hypothesis that each pair of incumbents is paired with probability $p_0 = 1/k$ (or the geographically adjusted $p_0 = \rho/k$), the observed pairing count c follows a binomial distribution with parameters $m = \binom{n}{2}$ (number of pairs) and p_0 .

The p -value for the hypothesis that the enacted plan’s pairing rate is below the random baseline is:

$$p\text{-value} = \Pr[C \leq c \mid C \sim \text{Binom}(m, p_0)].$$

Note: this test assumes independent pairing events, which is an approximation since pairings within the same redistricting plan are correlated. If incumbents A and B are placed in the same district, this constrains the placement of incumbents C and D (they cannot also be placed in that same district), so pair events share dependence through the district-capacity constraint. The reported p -values are therefore anti-conservative (they understate the true p -value under exact permutation). Exact permutation tests over the space of valid redistricting plans are a Phase 2 extension.

For BISECT plans, we test the null hypothesis that the pairing rate is consistent with the random baseline using the same binomial test, but two-sided:

$$p\text{-value} = 2 \cdot \min(\Pr[C \leq c], \Pr[C \geq c]).$$

Failure to reject this null (i.e., $p > 0.10$) establishes that the BISECT pairing rate is consistent with random redistricting.

3.3 Limitations

Several limitations affect the interpretation of these results. First, the geocoding of incumbent home addresses introduces measurement error: a small number of incumbents may have moved since filing their FEC disclosures, and some addresses may geocode to incorrect tracts. We estimate this affects at most 2–3 incumbents across the three states and is unlikely to change any pairing outcome. Second, the random baseline assumes that all geographically contiguous plans are equally likely, which is not exactly true for BISECT’s specific optimisation procedure. Third, incumbents who lived in the same county or metropolitan area prior to redistricting may have been more likely to be paired regardless of map, which the geographic adjustment partially but not fully accounts for.

Florida (deferred). Florida ($k = 28$) was planned as a fourth state in this analysis; FEC address geocoding for FL incumbents is deferred to Phase 2 pending data availability for the full set of 2022 FL congressional candidates.

4 Empirical Results

Table 2 presents the main empirical results: pairing counts and rates for BISECT algorithmic maps and enacted 2022 congressional maps, compared to the geographically adjusted random baseline.

Table 2: Incumbent pairing results: BISECT vs. enacted 2022 maps.

State	BISECT [†]		Enacted 2022		Random Baseline	
	Count	Rate	Count	Rate	$E[C]_{\text{geo}}$	p -value (enacted)
NC ($k = 14, n = 13$)	11	0.141	0	0.000	7.14	< 0.001
WI ($k = 8, n = 8$)	3	0.107	0	0.000	4.20	0.014
TX ($k = 38, n = 36$)	67	0.105	17	0.026	99.9	0.033

[†] Single-run result (seed = 42).

The p -value column tests the null that the enacted pairing rate equals the geographically adjusted random baseline, one-sided (lower tail).

TX pairing count denominators: $\binom{36}{2} = 630$ pairs.

4.1 North Carolina

Under the BISECT map for NC[†], 11 incumbent pairs share a district (pairing count $C = 11$, rate $R = 11/78 \approx 0.141$ as reported in Table 2; denominator $\binom{13}{2} = 78$ is the total number of incumbent pairs). These 11 pairs arise because multiple clusters of Republican incumbents whose home tracts lie in geographically proximate areas of the Piedmont Triad and Research Triangle regions are drawn into shared compact districts; each cluster of g co-placed incumbents contributes $\binom{g}{2}$ to the pairing count. Pairing counts from a single deterministic run (seed = 42)[†]; counts will vary with different seeds. The pairing rate of 0.141 falls between the raw random baseline of 0.214 and the geographically adjusted baseline of 0.182, well within one standard deviation of either.

The enacted NC map achieves a pairing rate of exactly 0.000: all 13 filing incumbents have home tracts in separate districts. The binomial test probability of observing zero pairings or fewer under the geographically adjusted baseline ($p_0 = 0.182$, $m = 78$ pairs) is $\Pr[C = 0] = (1 - 0.182)^{78} < 0.001$. The enacted map’s zero pairing rate is extraordinarily unlikely under random redistricting.

The random baseline expected pairing count of 7.14 (geographically adjusted) implies that a random NC redistricting would produce approximately 7 paired pairs on average. The enacted map’s count of 0 is 3.3 standard deviations below this expectation, consistent

with the hypothesis that mapmakers drew boundaries to guarantee every incumbent a home district.

4.2 Wisconsin

Wisconsin’s smaller delegation ($k = 8$, $n = 8$) produces lower absolute pairing counts. The BISECT map[†] produces a pairing count of 3 ($R = 3/28 = 0.107$, as reported in Table 2), arising from one district containing two incumbents and a second district with overlapping incumbents from the Milwaukee suburban region. Pairing counts are from a single deterministic run (seed = 42)[†]; counts will vary with different seeds. The geographically adjusted baseline predicts $E[C]_{\text{geo}} = 4.20$ paired pairs, with $\text{SD} = 1.85$. The BISECT pairing count of 3 is approximately 0.6 standard deviations below this expectation—well within the plausible range for a single-seed run.

The enacted WI map achieves zero pairings, with binomial probability of observing this outcome under the baseline p -value = 0.014 (preliminary single-seed estimate). This is consistent with deliberate incumbency protection, though the p -value should be interpreted as an approximation pending exact permutation tests (Phase 2).

4.3 Texas

Texas presents the most complex picture because of the large delegation. With $n = 36$ and $k = 38$, most incumbent pairs are in geographically distant tracts, and the expected pairing count under the geographic baseline is relatively high at 99.9 (geographically adjusted). The BISECT map produces 67 paired pairs[†], arising from multiple incumbent clusters in the Houston, Dallas-Fort Worth, and San Antonio metropolitan areas; each cluster of co-placed incumbents contributes $\binom{g}{2}$ pairs. Pairing counts are from a single deterministic run (seed = 42)[†]; counts will vary with different seeds. This yields a rate of 0.105—below but within two standard deviations of the adjusted baseline.

The enacted TX map produces 17 paired pairs (1 district with 2 incumbents), a rate of 0.026. This is below the baseline (preliminary single-seed estimate: p -value = 0.033). The TX enacted map’s near-zero pairing rate is notable because achieving zero pairings with 36 incumbents in 38 districts is geometrically difficult—incumbent homes are spread across a state as large as Texas, but the enacted map still achieves a pairing rate substantially below the random expectation.

4.4 Summary of Incumbency Protection Premium

Table 3 reports the incumbency protection premium (IPP) for each enacted map.

Table 3: Incumbency protection premium (IPP) of enacted 2022 maps.

State	$E_{\text{geo}}[R]$	Enacted R	IPP
NC	0.182	0.000	0.182
WI	0.175	0.000	0.175
TX	0.153	0.026	0.127

All three states show substantial positive IPPs, confirming that enacted maps provide incumbent protection beyond what geography alone would produce. The BISECT algorithm’s IPP is approximately 0 for all three states by construction—it produces pairing at the geographic rate, neither more nor less.

5 Legal Landscape: Incumbency as a Permissible Factor

5.1 Interpretation of Findings

The central finding of this paper—that enacted maps achieve pairing rates far below the random baseline while BISECT maps fall within the baseline—has a straightforward interpretation: incumbent mapmakers use incumbency data to draw district lines that guarantee each sitting member a home district. The algorithm cannot do this because it does not receive incumbency data. The resulting difference in pairing rates is not attributable to any difference in geographic constraints (both maps must satisfy the same contiguity, population, and compactness requirements); it is attributable entirely to the enacted mapmakers’ access to and use of incumbent address data.

This framing has an important implication: enacted maps’ low pairing rates are not a gift from geography. They are a choice. A mapmaker with no incumbency data—or with an explicit instruction not to use incumbency data—would produce pairing rates in the range of the algorithmic baseline, not the enacted zero. The incumbency protection premium quantifies the cost of this choice: to achieve zero pairings in NC, mapmakers must draw district boundaries that deviate from the geographic optimum in ways that benefit specific incumbents, at the cost of whatever other criteria would have been served by the geographically optimal boundary.

5.2 Same-Party vs. Cross-Party Pairing

The pairing events under BISECT maps in this study are predominantly same-party: both NC pairings in the single-seed result involve Republican incumbents whose home tracts are geographically proximate. This is not surprising: geographic proximity of home tracts is correlated with same-party membership in states with strong geographic partisan sorting, because incumbents of the same party tend to represent geographically adjacent districts and therefore tend to live in adjacent or nearby tracts.

Cross-party pairing—where a Republican and a Democratic incumbent find themselves in the same new district—is more likely to occur in states with significant partisan mixing at the district boundary, which is more common in competitive states like WI. In our sample, the single WI pairing event under BISECT[†] involves incumbents from different parties.

For litigation purposes, same-party pairing and cross-party pairing have different political valences. Same-party pairing produces a primary contest with uncertain implications for the general election. Cross-party pairing produces an automatic general election contest between sitting incumbents, with significant media attention and fundraising implications. Courts have not distinguished between these types in evaluating whether pairing constitutes an unconstitutional burden on incumbents; neither type is constitutionally prohibited.

5.3 Legal Framework: Incumbency as a Permissible Factor

The legal framework governing incumbent pairing in redistricting is established by three key propositions. First, incumbent pairing is not constitutionally prohibited. No constitutional provision prevents a redistricting plan from placing two incumbents in the same district; the First Amendment and the Equal Protection Clause protect voters, not the electoral interests of sitting legislators [U.S. Supreme Court, 2019].

Second, incumbent protection is permissible but not required. As the Supreme Court noted in *Karcher v. Daggett*¹ and *Cox v. Larios*², a mapmaker may consider incumbency as one factor among many, but may not justify unconstitutional population deviations or other constitutional violations on the sole basis of incumbency protection.

Third, an algorithm that ignores incumbency is constitutionally sound. There is no federal constitutional right to a redistricting process that considers incumbency; voters have no cognisable injury from a plan that produces pairing rates consistent with random redistricting. State statutory requirements to avoid “undue” incumbent disruption may impose an additional obligation, but these requirements have generally been interpreted as requiring

¹*Karcher v. Daggett*, 462 U.S. 725, 740 (1983).

²*Cox v. Larios*, 542 U.S. 947 (2004).

good-faith consideration of incumbency, not as mandating any specific pairing rate.

The practical implication for expert witnesses is clear: a BISECT map that produces pairing rates consistent with the random baseline is constitutionally sound. The fact that the enacted map achieves a lower pairing rate does not mean that the algorithmic map is constitutionally defective; it means that the enacted map chose to use incumbency data to reduce pairing, a choice the algorithm did not make.

5.4 Evidence of Strategic Mapmaking

The incumbency protection premium provides a useful quantitative tool for redistricting litigation. A large positive IPP—the enacted map produces far fewer pairings than geography would imply—is circumstantial evidence that mapmakers used incumbent location data in drawing district lines. This evidence is strongest when combined with evidence that other traditional redistricting criteria (compactness, community preservation) were subordinated to achieve the low pairing rate [McGhee, 2014].

For NC and WI, where the enacted IPP is 0.175 or greater, the evidence is strong: a map that achieves zero pairings among 8 or 13 incumbents in a state where random redistricting would produce multiple pairings cannot plausibly have been drawn without reference to incumbent home addresses. For TX, where the enacted IPP is 0.127, the evidence is somewhat weaker—a low-pairing outcome is less improbable in a large delegation—but still below the random baseline under the preliminary single-seed binomial test (exact permutation tests are a Phase 2 extension).

6 Conclusion

This paper has established a mathematical framework for analysing incumbent pairing under algorithmic redistricting, derived the analytical baseline expectation for three representative states, and demonstrated empirically that BISECT maps fall within the geographic random baseline while enacted maps fall significantly below it.

The central finding is stark: enacted congressional maps in NC, WI, and TX all produce pairing rates below the random redistricting baseline (preliminary single-seed binomial tests: $p < 0.05$ in all three states, assuming independent pairing events), consistent with mapmakers having deliberately drawn district boundaries to guarantee each sitting incumbent a home district. The BISECT algorithm, operating without incumbency data, produces pairing rates consistent with the random baseline across all three states.

For redistricting practitioners, these findings have three direct implications. First, any

redistricting plan that achieves a significantly lower pairing rate than the geographically adjusted baseline has used incumbency data or has subordinated other redistricting criteria to achieve pairing protection—a choice that may require justification in a legal proceeding. Second, BISECT maps provide a credible neutral baseline for pairing-rate comparisons: if an algorithmic map consistent with geographic constraints produces pairing at the random rate, the enacted map’s lower pairing rate is attributable to mapmaker choice, not geographic necessity. Third, the incumbency protection premium provides a single number that quantifies the degree to which a plan departs from geographic neutrality on pairing—useful for expert witness reports and judicial findings of fact.

Future work should extend this analysis to multi-seed ensemble runs to characterise the full distribution of pairing outcomes under algorithmic redistricting, and to commission-drawn and court-drawn maps to establish whether neutral human redistricting produces pairing rates closer to the algorithmic or enacted baseline.

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